Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

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Abstract

This dissertation presents an explainable (XAI-first) financial-alert system for retail investors in the US equity market (NYSE and NASDAQ). The system raises Telegram alerts from two independent triggers — abrupt market movements, detected as statistical anomalies, and incoming financial news — and, for every alert, exposes a fully traceable explanation: the detected event, the reasoning applied, the sources, and historical precedents. Its core is a news–market correlation engine that, given a news item, retrieves analogous historical news from a knowledge base and measures the impact those precedents had on prices, in the manner of an event study and without lookahead. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components — sentence-embedding retrieval, statistical anomaly detection and transparent explanation — in a functional, explainable and reproducible system. On real data, semantic retrieval clearly outperformed lexical and trivial baselines (cross-ticker precision@5 of 0.55 against 0.24 for random retrieval), and the volatility-normalised detector fired far more consistently across stocks than a fixed threshold. The results are honest and reproducible, with limitations and future work stated explicitly.

Keywords: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Resumo

Esta dissertação apresenta um sistema de alertas financeiros explicável (XAI-first) para investidores de retalho no mercado acionista dos Estados Unidos (NYSE e NASDAQ). O sistema envia alertas via Telegram a partir de dois gatilhos independentes — movimentos abruptos de mercado, detetados como anomalias estatísticas, e novas notícias financeiras — e, para cada alerta, expõe uma explicação totalmente rastreável: o evento detetado, o raciocínio aplicado, as fontes e os precedentes históricos. O seu núcleo é um motor de correlação notícia–mercado que, dada uma notícia, recupera notícias históricas análogas de uma base de conhecimento e mede o impacto que esses precedentes tiveram nos preços, à maneira de um estudo de evento e sem lookahead. Enquanto contribuição de Engenharia de Inteligência Artificial, o trabalho integra, aplica e avalia criticamente componentes existentes — recuperação por embeddings de frases, deteção estatística de anomalias e explicação transparente — num sistema funcional, explicável e reprodutível. Em dados reais, a recuperação semântica superou claramente as baselines lexical e triviais (precision@5 cross-ticker de 0,55 face a 0,24 do acaso) e o detetor normalizado pela volatilidade disparou de forma muito mais consistente entre ações do que um limiar fixo. Os resultados são honestos e reprodutíveis, com limitações e trabalho futuro explícitos.

Palavras-chave: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

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List of Symbols

r_t	log-return of an asset on day t	—
μ_t	rolling mean of returns	—
σ_t	rolling standard deviation of returns	—
z_t	z-score of the return on day t	—
k	anomaly threshold (in standard deviations)	—
w	rolling-window length	d

List of Acronyms

AI	Artificial Intelligence.
NASDAQ	National Association of Securities Dealers Automated Quotations.
NYSE	New York Stock Exchange.
RQ	Research Question.
SIFMA	Securities Industry and Financial Markets Association.

Chapter 1

Introduction

This chapter introduces the dissertation. It first sets the context — the scale of the United States equity market, the growth of retail investing, and the rapid yet opaque adoption of artificial intelligence in finance — and then states the problem the work addresses, the objectives and research questions that guide it, the scientific contributions it claims, and the structure of the remaining document.

1.1 Contextualization

The United States equity market is the largest and among the deepest and most liquid in the world. According to the Securities Industry and Financial Markets Association (SIFMA) Capital Markets Fact Book, US equity markets represented 49.1% of the US$126.7 trillion in global equity market capitalisation at the end of 2024 — about US$62.2 trillion, some 5.3 times the size of the next-largest market, China (Securities Industry and Financial Markets Association (SIFMA) 2025). The two principal trading venues are the New York Stock Exchange (NYSE) and National Association of Securities Dealers Automated Quotations (NASDAQ), which together list the large-capitalisation companies that dominate global indices. As shown in Figure 1.1, this market has grown markedly over the last decade, from US$25.1 trillion in 2015 to US$62.2 trillion in 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025) — a scale and liquidity that make it the natural focus for a retail-oriented alerting system.

Equity ownership is also widespread among US households. In 2025, 62% of Americans reported owning stock — whether individual shares or holdings within mutual funds and retirement accounts — matching the 2024 figure and continuing a rebound after more than a decade below that level (Gallup 2025). Ownership is unevenly distributed, rising to 87% among households earning US$100,000 or more and falling to 28% among those earning under US$50,000 (Gallup 2025). Beyond ownership, non-professional ("retail") investors have become a material part of day-to-day trading, helped by commission-free mobile brokerages. This combination of broad ownership and growing direct participation defines the audience this work serves: the non-professional investor who must make sense of the market without the tools available to institutions.

In parallel, artificial intelligence has been adopted rapidly across financial services. A 2026 industry survey by the Cambridge Centre for Alternative Finance found that 81% of surveyed financial-services firms were adopting Artificial Intelligence (AI) at some level, 40% reported advanced adoption, and 71% were already using generative AI (Cambridge Centre for Alternative Finance 2026). Yet many high-performing models behave as opaque "black boxes", offering little insight into how a given output was produced (Adadi and Berrada

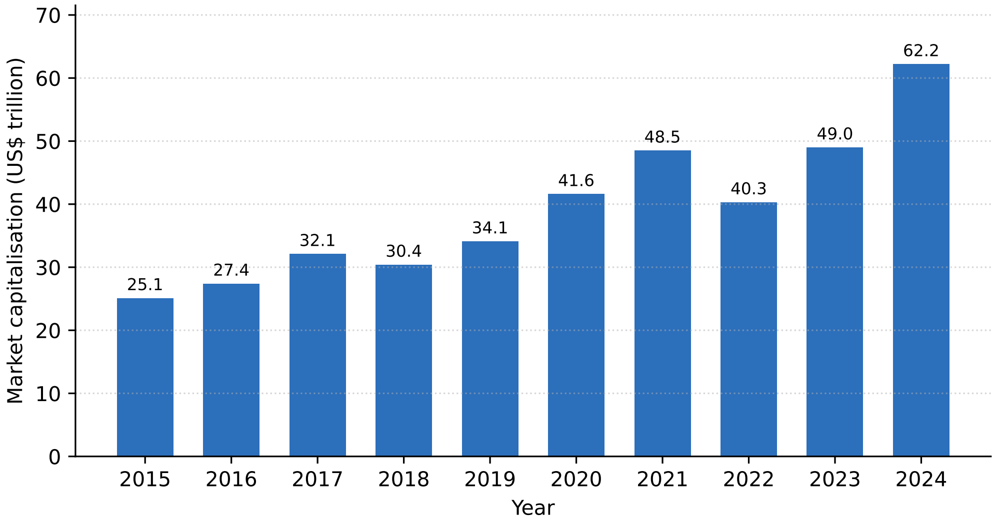

Figure 1.1: US equity market capitalisation, 2015–2024 (US$ trillion). Figure generated from data reported in the SIFMA 2025 Capital Markets Fact Book (Securities Industry and Financial Markets Association (SIFMA) 2025).

2018; Barredo Arrieta et al. 2020). For a retail investor, who must act on — and bear the consequences of — a system's output, this lack of transparency is not a minor inconvenience but a barrier to trust and to informed action. It is the principle that guides the present work.

1.2 Problem Statement

A retail investor who wishes to stay informed faces two simultaneous streams of information: continuous price movements across thousands of US tickers, and a constant flow of financial news. Two distinct problems follow. First, distinguishing a genuinely abnormal price movement from ordinary volatility requires statistical judgement and constant attention. Second, when a relevant news item appears, assessing its likely impact requires knowledge of how analogous past events affected prices — information that is rarely at hand. Existing consumer tools tend to deliver alerts without explanation, or rely on models whose internal logic is hidden, which is precisely the transparency problem that explainable AI seeks to address (Adadi and Berrada 2018; Barredo Arrieta et al. 2020).

The problem this work tackles is therefore not price prediction, but *explanation*: detecting abrupt market movements and relevant news, and presenting, for each alert, a traceable account of what was detected, the reasoning applied, the sources consulted, and concrete historical precedents of similar events together with their observed impact.

1.3 Research Questions and Objectives

The general objective is to design, implement and evaluate an explainable, reproducible financial-alert system for US retail investors, driven by two independent triggers (abrupt market movements and incoming news) and delivered through Telegram. This is pursued through three research questions:

- **Research Question (RQ)1.** Can transparent statistical methods detect abrupt market movements with enough consistency to be useful to a retail investor, while remaining explainable?
- **RQ2.** Can a news–market correlation engine retrieve genuinely analogous historical news and quantify their observed price impact, without lookahead, better than trivial baselines?
- **RQ3.** Can the resulting explanations be made faithful to the system's actual logic and useful to a retail investor?

The supporting objectives are: (i) build a thin end-to-end slice early to control integration risk; (ii) develop each component in its simplest defensible form; (iii) evaluate each component against a design defined beforehand (Chapter 5); and (iv) keep the whole system reproducible and transparent.

1.4 Scientific Contributions

This is an Artificial Intelligence *Engineering* dissertation. The contribution is not to invent new algorithms but to *integrate, apply and critically evaluate* existing components in a functional, explainable and reproducible system. Concretely, the contributions are:

- A documented, reproducible methodology for **news–market impact correlation**, combining sentence-embedding retrieval with event-study impact windows, with explicit choices of similarity metric and window and explicit avoidance of lookahead.
- **CLARION**, an **XAI-first alerting system** that exposes its full reasoning chain (detection, evidence, sources, precedents) to a non-expert user and is faithful to its own computation by construction.
- A **critical evaluation** of each component against baselines and a transparent design, on real data, reporting honest results and limitations.

1.5 Document Structure

The remainder of the document is organised as follows. Chapter 2 reviews the state of the art on anomaly detection, explainable AI, financial natural-language processing and event-study approaches to news–market impact, with comparative tables and a positioning of this work. Chapter 3 describes the methods, datasets and materials used, together with the evaluation methodology and the responsible-AI considerations. Chapter 4 presents CLARION, the proposed system, at the level of its architecture and design decisions. Chapter 5 reports the evaluation as a set of case studies on real data. Finally, Chapter 6 draws the main conclusions, revisits the research questions and contributions, and states the limitations and future work.

Chapter 2

State of the Art

This chapter reviews the body of work on which the dissertation builds. It opens with the behaviour of the retail investor and the problem of information overload, which motivate the system, and then surveys, in turn, anomaly detection in financial markets, explainable artificial intelligence, financial natural-language processing and text representation, and event-study approaches to news–market impact. Each area combines seminal and recent contributions, and comparative tables summarise the approaches, their mechanisms and their limitations. The chapter closes by positioning the present work against the alternatives and identifying the gap it addresses.

2.1 Retail Investing and Information Overload

The audience of this work is the non-professional, or "retail", investor. Understanding how such investors consume information and react to news is therefore the natural starting point, because it both motivates the system and constrains what a useful alert should contain.

A central and durable finding of behavioural finance is that individual investors are strongly influenced by *attention*. In their seminal study, Barber and Odean (2008) show that individual investors are net buyers of attention-grabbing stocks — those in the news, those with unusually high trading volume, and those with extreme one-day returns — whereas they do not face the same search problem when selling, because they tend to sell only what they already own. The explanation offered is economic: faced with thousands of stocks they could potentially buy, investors fall back on the small subset that has captured their attention. This asymmetry matters directly for an alerting system: the very events that draw a retail investor's attention — a sharp price move or a salient news item — are precisely the two triggers around which this work is built, and an alert that adds context to such an event is intervening exactly where attention-driven decisions are made.

The role of the media in shaping investor behaviour has likewise been quantified. Tetlock (2007) constructs a daily pessimism measure from a popular Wall Street Journal column and finds that high media pessimism predicts downward pressure on prices followed by a reversion to fundamentals, and that unusually high or low pessimism predicts elevated trading volume. The study is doubly relevant here: it establishes that news content carries measurable, market-moving information, and it is an early demonstration that *textual* signals can be extracted and related to market outcomes — the same premise that underlies the news–market correlation engine of this dissertation.

The composition and weight of retail participation have also shifted markedly in recent years, driven by commission-free mobile brokerages. Studying a large sample of users of one such platform, Welch (2022) documents that this new cohort of investors increased its holdings during the March 2020 bear market rather than panicking, and that an aggregated "crowd consensus" portfolio exhibited both reasonable timing and positive performance over the period studied. Whatever the debate over retail investors' sophistication, the empirical picture is one of a large, active and consequential population — consistent with the broad ownership figures reported by Gallup (2025) and the scale of the market documented by Securities Industry and Financial Markets Association (SIFMA) (2025) in Chapter 1.

Taken together, this literature frames the problem the system addresses. Retail investors are attention-constrained, react to news, and now participate in large numbers, yet they must do so under a continuous flood of price movements and headlines that they cannot individually process. The response pursued here is not to predict the market for them, but to reduce the cost of *interpreting* the events that already command their attention, by attaching to each alert a transparent explanation and concrete historical evidence.

2.2 Anomaly Detection in Financial Markets

The first trigger of the system rests on detecting abnormal price movements, which places it within the mature field of anomaly detection. The survey of Chandola, Banerjee, and Kumar (2009) provides the standard reference and taxonomy, distinguishing *point* anomalies (an individual observation that is abnormal with respect to the rest), *contextual* anomalies (abnormal only in a particular context, such as a time of year), and *collective* anomalies (a set of observations that is jointly abnormal), and grouping detection methods into statistical, distance- and density-based, and machine-learning families. A sharp single-day return is naturally treated as a contextual point anomaly: abnormal relative to the asset's own recent behaviour. Figure 2.1 summarises the families relevant to this work.

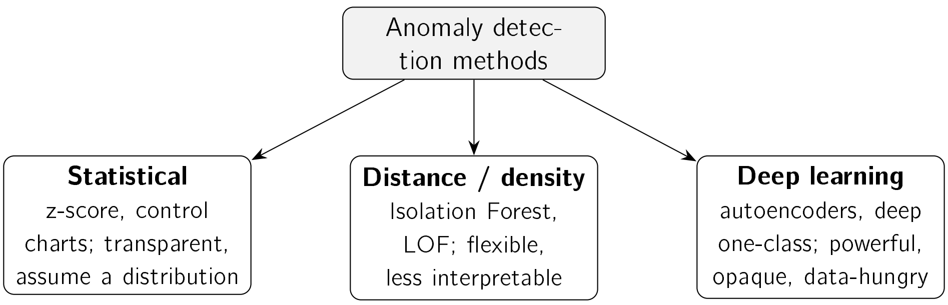

Figure 2.1: Families of anomaly-detection methods relevant to this work, following Chandola, Banerjee, and Kumar (2009) and Pang et al. (2021). Interpretability decreases, and modelling capacity increases, from left to right.

For financial returns, simple statistical methods — flagging deviations from a rolling mean and standard deviation, that is, a z-score — are attractive precisely because they are transparent and easy to justify, an advantage for an explainable system. They do, however, assume a locally stable distribution, an assumption that the rolling estimation window is designed to accommodate. More expressive unsupervised detectors relax such assumptions: Isolation Forest (Liu, Ting, and Zhou 2008) isolates anomalies through random partitioning, requiring fewer splits to separate an outlier, and scales well to large, high-dimensional data. The cost is reduced direct interpretability, since the anomaly score is an ensemble property rather than a quantity the user can read off. At the far end of the spectrum, the review by

Pang et al. (2021) surveys deep-learning detectors — autoencoders, deep one-class models and others — which can capture rich, non-linear structure but are data-hungry and largely opaque, and whose advantage is most pronounced in high-dimensional settings such as images or complex multivariate streams rather than in a single return series. Table 2.1 compares the three families.

Table 2.1: Anomaly-detection approaches relevant to this work.

Approach	**How it works**	**Strengths / limitations**
Statistical z-score (rolling) (Chandola, Banerjee, and Kumar 2009)	Flags returns that deviate from a rolling mean and standard deviation	Transparent and easy to explain; assumes a locally stable distribution
Isolation Forest (Liu, Ting, and Zhou 2008)	Isolates points via random partitioning (ensemble)	Scalable, handles high dimensionality; score is not directly interpretable
Deep detectors (Pang et al. 2021)	Learn a model of normality (e.g. autoencoders)	High capacity for complex data; opaque, data-hungry, hard to justify

The trade-off this table makes explicit — capacity against interpretability — is the one that governs the design choice in Chapter 3. Because the system must *explain* every alert to a non-expert, and because the signal is a single, low-dimensional return series rather than a high-dimensional stream, the transparent statistical detector is preferred, with the more expressive families noted as alternatives whose opacity is not warranted by the problem at hand.

2.3 Explainable Artificial Intelligence

As AI systems are deployed in consequential settings, the inability to inspect their reasoning has become a central concern, and explainable AI (XAI) is the field that addresses it. The surveys of Adadi and Berrada (2018) and Barredo Arrieta et al. (2020) map the area: its motivations (trust, accountability, debugging, regulatory compliance), a taxonomy that separates models that are *transparent* (interpretable in their own right) from *post-hoc* methods that explain an already-trained black box, and the open challenges on the path to responsible AI. The earlier survey of Guidotti et al. (2018) organises the post-hoc literature specifically by the problem being solved — explaining a whole model, explaining a single outcome, or inspecting the internals — which is a useful lens for situating the techniques most relevant here. Figure 2.2 sketches this division.

Among post-hoc, model-agnostic techniques, two are widely used. LIME (Ribeiro, Singh, and Guestrin 2016) explains a single prediction by fitting a simple, interpretable surrogate model in the local neighbourhood of the instance; it is intuitive and broadly applicable, but its explanations can be unstable under resampling and are only locally faithful. SHAP (Lundberg and Lee 2017) attributes a prediction to its input features using Shapley values from cooperative game theory, which gives it desirable consistency properties and a unifying theoretical grounding, at a higher computational cost. Both are valuable when one is committed to a black-box model and needs to interrogate it after the fact.

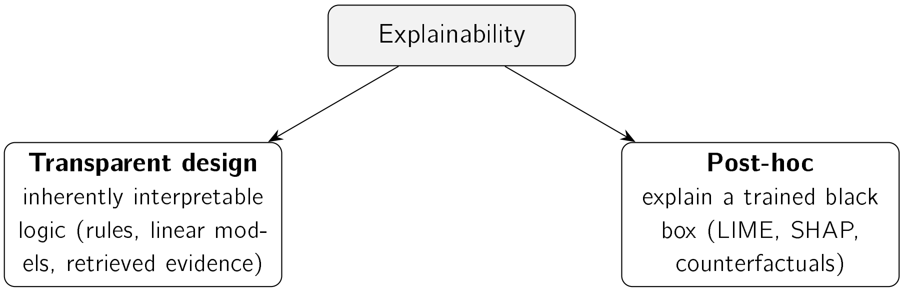

Figure 2.2: The two broad routes to explainability (Barredo Arrieta et al. 2020; Guidotti et al. 2018): building models that are transparent by design, or explaining a black box after the fact. This work follows the former.

A strand of the literature questions that commitment directly. Rudin (2019) argues forcefully that, for high-stakes decisions, one should *stop explaining* black-box models and instead use models that are interpretable in the first place, on the grounds that post-hoc explanations are approximations that can mislead and can entrench poor practice. This position is particularly pertinent to a tool that a non-expert will act upon financially: an explanation that is itself an approximation of an opaque model carries a risk that a transparent design avoids by construction. The present work takes this view as a guiding principle, pursuing explainability primarily through transparent design — a statistical detector whose every quantity is exposed, and retrieved historical precedents shown as direct evidence — and treating feature attribution as an optional complement rather than the foundation.

Finally, the field has recognised that explanations must themselves be evaluated rather than assumed. Doshi-Velez and Kim (2017) propose a taxonomy for the rigorous evaluation of interpretability, distinguishing application-grounded, human-grounded and functionally-grounded assessments, and emphasising *fidelity* — the degree to which an explanation reflects the system's actual computation — as a core criterion. This framing is adopted in Chapter 5, where the explanation is assessed for faithfulness by construction, with the (human-grounded) question of usefulness stated openly as a limitation. Table 2.2 summarises the approaches considered.

Table 2.2: Explainability approaches considered.

Method	**Type**	**Strengths / limitations**
LIME (Ribeiro, Singh, and Guestrin 2016)	Local surrogate, post-hoc	Intuitive, model-agnostic; explanations can be unstable, only locally faithful
SHAP (Lundberg and Lee 2017)	Shapley-value attribution, post-hoc	Consistent, theoretically grounded; computationally costly
Transparent design (Barredo Arrieta et al. 2020; Rudin 2019)	Inherently interpretable logic	No approximation, faithful by construction; constrains model choice

2.4 Financial NLP and Text Representation

The news trigger depends on comparing the meaning of news items, which requires representing text numerically. The literature on financial natural-language processing spans three broad generations: hand-crafted lexicons, learned word embeddings, and contextual neural language models.

The earliest and still-influential approach in finance is the domain dictionary. Loughran and McDonald (2011) show that general-purpose sentiment word lists misclassify ordinary financial vocabulary — words such as "liability" or "tax" that are negative in everyday language but neutral in a financial report — and develop finance-specific word lists that better capture tone in regulatory filings, relating them to returns, volume and volatility. Lexicon methods are fully transparent and remain a strong baseline, but they are bounded by their vocabulary and cannot capture meaning that is not expressed by listed words.

Learned embeddings address this limitation by placing words in a continuous vector space where proximity reflects distributional similarity. Word2vec (Mikolov et al. 2013) learns such vectors efficiently from local context windows, and GloVe (Pennington, Socher, and Manning 2014) obtains comparable representations from global word–word co-occurrence statistics. Both, however, assign a single vector to each word irrespective of context, so they cannot distinguish the different senses a word takes in different sentences. The Transformer architecture (Vaswani et al. 2017), built entirely on attention mechanisms and dispensing with recurrence, removed this limitation and reshaped the field: BERT (Devlin et al. 2019) uses it to produce deeply *contextual* token representations that substantially advanced language understanding. Using BERT directly to compare two texts is nonetheless expensive, because it processes sentence pairs jointly; Sentence-BERT (Reimers and Gurevych 2019) resolves this by producing a single fixed-length embedding per sentence that can be compared with a cheap cosine similarity — exactly the property on which precedent retrieval in this work relies.

Two further developments are relevant. First, domain adaptation: FinBERT models fine-tune BERT on financial text, for sentiment classification (Araci 2019) and for broader financial communications (Yang, Uy, and Huang 2020), improving on general-purpose models within the domain. Second, the recent rise of large language models, exemplified in finance by BloombergGPT (Wu et al. 2023), a fifty-billion-parameter model trained on a large financial corpus that outperforms comparable general models on financial tasks. Such models are powerful but costly to run and opaque, and they fall outside this work's constraint of free, transparent and reproducible components; they are noted as part of the landscape and as future work rather than as a chosen tool. Table 2.3 compares the representations.

The progression in Table 2.3 clarifies the design choice. Sentence-BERT occupies the sweet spot for this work: it captures contextual meaning well beyond lexicons and static embeddings, yet, unlike raw BERT or large language models, it yields directly comparable sentence vectors at low cost, which keeps the retrieval step both reproducible and free to run.

2.5 Event Studies and News–Market Impact

Having retrieved analogous past news, the system must quantify the impact those events had, which is the province of the *event study*. The methodology originates with Fama et al.

Table 2.3: Text-representation approaches for financial news.

Approach	Representation	Strengths / limitations
Finance lexicons (Loughran and McDonald 2011)	Curated word lists / tone	Fully transparent; bounded by vocabulary
word2vec / GloVe (Mikolov et al. 2013; Pennington, Socher, and Manning 2014)	Static word vectors	Efficient; one vector per word, no context
BERT (Devlin et al. 2019; Vaswani et al. 2017)	Contextual token embeddings	Strong understanding; pairwise similarity is costly
Sentence-BERT (Reimers and Gurevych 2019)	Sentence embeddings (siamese)	Efficient cosine similarity; the choice in this work
FinBERT (Araci 2019; Yang, Uy, and Huang 2020)	Domain-adapted language model	Finance-tuned; oriented to sentiment / communications
Financial LLMs (Wu et al. 2023)	Large generative model	Very capable; costly, opaque, outside free-tier scope

(1969), whose study of stock-price adjustment to new information introduced the event-study design and provided an early test of market efficiency by measuring returns around an event date. Brown and Warner (1985) subsequently established the now-standard practice for event studies based on daily returns, characterising how abnormal returns should be measured and tested over short post-event windows — the basis for the +1, +3 and +5-day horizons used in this work. Crucially for the honesty of the present design, the event-study impact is measured strictly *after* the event, so it characterises an outcome rather than forecasting one.

Relating *news text* to such impact is the subject of a growing literature, of which Tetlock (2007) is an early and influential example, demonstrating a measurable link between media tone and subsequent market behaviour. Conducting such an analysis historically and at scale requires news aligned to prices, which is the contribution of the FNSPID dataset (Dong, Fan, and Peng 2024): financial news time-aligned with daily prices for thousands of companies, enabling a news-to-impact knowledge base without bespoke scraping. Its limitations are those of any corpus — coverage and time period — and are documented in the data card of Chapter 3. The combination of a semantic retrieval step with an event-study impact measurement over such a corpus is the methodological core of this dissertation.

2.6 Comparative Analysis and Positioning

Across these areas, the literature offers strong individual components but few integrated, explainable, retail-facing systems that combine them. The choices made in this work, and the alternatives against which they were weighed, are summarised in Table 2.4. The guiding principle is *defensible simplicity*, aligned with the interpretable-by-design argument of Rudin

(2019): where two options perform comparably, the more transparent and easier-to-explain one is preferred, because the system's purpose is to be understood and acted upon by a non-expert.

Table 2.4: Component choices in this work versus alternatives.

Component	**Chosen**	**Alternative**	**Why**
Anomaly detection	*z*-score (rolling)	Isolation Forest, deep detectors	Transparency; low-dimensional signal
News retrieval	Sentence-BERT + cosine	Lexicons, word2vec, LLMs	Contextual yet cheap and reproducible
Impact measurement	Event-study windows	—	Standard, reproducible, no forecasting
Explanation	Transparent design + precedents	LIME / SHAP only	Faithful by construction

This positioning also clarifies the nature of the contribution. None of the individual building blocks is novel; what is comparatively scarce, and what this work provides, is their disciplined integration into a single explainable system for the retail investor, with each choice justified against the alternatives reviewed above.

2.7 Chapter Conclusions

This chapter reviewed the foundations of the work across four areas. The behavioural-finance literature established that retail investors are attention- and news-driven, motivating an explainable alerting tool rather than a predictive one. The anomaly-detection literature supplied a clear capacity–interpretability trade-off that favours a transparent statistical detector for a low-dimensional return series. The XAI literature, and in particular the interpretable-by-design argument, supported pursuing explainability through transparent design and assessing it by fidelity. The financial-NLP literature traced the path from lexicons to sentence embeddings and identified Sentence-BERT as the appropriate, reproducible choice for semantic retrieval, while event-study methodology and the FNSPID dataset provided the means to measure news impact honestly and at scale. What the literature offers in mature individual components, it largely lacks as an *integrated*, explainable, retail-facing system — the gap this dissertation addresses, and which the following chapters develop.

Chapter 3

Methods and Materials

This chapter describes the methods, materials and methodology used in the dissertation. It states the overall approach, the datasets and data sources, the models and techniques selected for each component (with their justification), the supporting tools, the responsible-AI considerations, and the evaluation methodology defined in advance. Throughout, the guiding principle is *defensible simplicity*: where options perform comparably, the more transparent and explainable one is preferred. The system that realises these methods is presented in Chapter 4, and the experiments that apply this methodology are reported in Chapter 5.

3.1 Approach and Methodology

The work follows an engineering methodology centred on incremental, end-to-end delivery. A thin slice of the system was built and proven end to end before breadth was added, so that integration risk was controlled early; each component was then developed in its simplest defensible form and only made more sophisticated where the added complexity was justified. This reflects the framing of the work as Artificial Intelligence *engineering*: the contribution lies in the integration, application and critical evaluation of existing components, not in the invention of new algorithms.

3.2 Datasets and Data Sources

The system uses two clearly separated data layers. The **historical** layer is built offline from FNSPID (Dong, Fan, and Peng 2024), which provides financial news already time-aligned with prices; for each historical news item the system stores its text, date, ticker, a sentence embedding, and the observed post-event price impact. The **live** layer provides, in real time, prices (from a free market-data service, with a second free provider as fallback) and news (from a free company-news service and from financial syndication feeds). Free news services are not used to build history because their free tiers are typically limited to recent windows and are delayed; the exact ticker subset, time window and preprocessing are documented in the data card.

3.3 Models and Techniques

3.3.1 Statistical Anomaly Detection

Abrupt movements are detected with a transparent statistical rule. Let r_t be the (log-)return of an asset on day t, and let μ_t and σ_t be the mean and standard deviation of returns over a trailing window of w days. The day is flagged as anomalous when the absolute z-score exceeds a threshold k:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \qquad \text{anomaly} \iff |z_t| > k. \tag{3.1}$$

This statistical family is standard (Chandola, Banerjee, and Kumar 2009) and is chosen for its transparency: the user can see exactly why an alert fired. More expressive detectors such as Isolation Forest (Liu, Ting, and Zhou 2008) were considered but not adopted as the default, because their reduced interpretability conflicts with the XAI-first goal; the window w and threshold k are documented design choices, evaluated in Chapter 5.

3.3.2 Semantic Retrieval and Impact Measurement

The news–market correlation engine is the core of the work. Each news item is encoded into a sentence embedding using a Sentence-BERT model (Reimers and Gurevych 2019), chosen because it allows efficient comparison of texts through cosine similarity. Given a new item, the engine retrieves the k most similar historical news items from the knowledge base and reports the impact those precedents had. Impact is measured in the manner of an event study (Brown and Warner 1985): the asset's return is computed over fixed post-event windows (e.g. $+1$, $+3$ days), measured strictly after the event so that no future information leaks into the evidence. The similarity metric and the impact windows are explicit design choices and form part of the contribution.

3.3.3 Explanation Techniques

Explanation is pursued primarily through transparent design, following the principles set out in the XAI literature (Adadi and Berrada 2018; Barredo Arrieta et al. 2020): an explanation the user can follow step by step. For every alert the system combines (i) the rule or measure that produced the detection (the z-score and its parameters), (ii) the retrieved historical precedents and their observed impact as evidence, and (iii), optionally, feature attribution over the detector using SHAP (Lundberg and Lee 2017). Financial sentiment from FinBERT (Araci 2019) is an optional enrichment that is cuttable if it does not add defensible value.

3.4 Tools and Frameworks

The system is implemented in the Python ecosystem, using established open-source libraries for numerical computing, market-data access, sentence embeddings and figure generation. The choice favours mature, well-documented and freely available tools, consistent with the project's constraint of using only free resources and with the goal of reproducibility. The concrete engineering — environment, repository organisation and testing — is described at design level in Chapter 4 and in Appendix A.

3.5 Responsible AI and Ethics

The work is bound by several responsible-AI commitments. The system performs no price prediction and gives no investment advice: it surfaces observed past outcomes as evidence and states this explicitly in every alert. All data sources are used within their free tiers and licences, with the FNSPID dataset attributed as required by its CC BY-SA 4.0 licence. No personal data is processed, and no secret or credential is stored in the project's versioned files.

3.6 Evaluation Methodology

The evaluation is defined in advance, to avoid fishing for favourable results, and is reported in Chapter 5. In summary: the anomaly detector is assessed by the consistency of its firing rate across instruments of differing volatility (a label-free argument) and by agreement with an explicit extreme-move proxy, with an ablation over the estimation window; the correlation engine is assessed on retrieval quality (cross-ticker precision@k with a sector relevance proxy, against random, recency and lexical baselines, over multiple random samples and two embedding models) and on the measured event-study impact; and the explanation engine is assessed for fidelity to the actual logic and, as a stated limitation, for usefulness. Because the system performs neither price prediction nor trading, profit-based metrics are explicitly out of scope.

Three guarantees apply throughout. First, **no lookahead**: features at any instant use only information available at or before that instant, and historical impact uses only post-event windows. Second, **reproducibility**: random seeds are fixed, dependencies are pinned, and every data and results figure is produced by a script in the repository. Third, **honesty**: only results that can be reproduced and explained are reported, and no data, metric or citation is fabricated.

3.7 Chapter Conclusions

This chapter set out the methodology of defensible simplicity, the two-layer data architecture, the models and techniques chosen for detection, retrieval, impact measurement and explanation, and the evaluation methodology and rigour guarantees fixed in advance. The next chapter shows how these methods are integrated into CLARION, the proposed system.

Chapter 4

CLARION: An Explainable Financial-Alert System for Retail Investors

This chapter presents Clarion, the system designed and built in this dissertation. Whereas Chapter 3 justified *what* methods were chosen and *why*, this chapter describes *how* they are integrated into a coherent system: its overall architecture, the historical knowledge base and correlation engine at its core, the anomaly trigger, the explanation engine that makes every alert traceable, and the delivery of alerts to the investor. Consistent with the framing of the work as AI engineering, the contribution lies in integrating existing models and techniques into a functional, explainable and reproducible whole.

4.1 Overview

Clarion watches the US equity market on behalf of a retail investor and raises an alert whenever one of two independent triggers fires: an abrupt price movement, or a relevant incoming news item. Each alert is accompanied by a full, traceable explanation rather than a bare notification. The system is organised as a pipeline of well-separated components, each with a single responsibility, connected by a common data schema so that the live and historical layers are directly comparable.

4.2 Architecture

The system is driven by the two triggers and converges on a single explanation step before delivery. Figure 4.1 shows the components and the flow of data between them. The live layer supplies real-time prices and news; the historical knowledge base supplies the precedents; the anomaly detector and the correlation engine produce the evidence; the explanation engine assembles a traceable account; and the delivery component sends the alert to the investor.

Three engineering principles recur throughout and make the system both testable and defensible. First, **a thin end-to-end slice was built before breadth**: the market-movement trigger was implemented and proven end to end — from price retrieval to a real alert — before the heavier correlation engine was added, reducing integration risk. Second, **pure logic is separated from input/output**, so the numerical core can be verified deterministically and offline, without a network connection or the heavy model stack. Third, **components are programmed against interfaces**: the embedding step is defined abstractly, with two

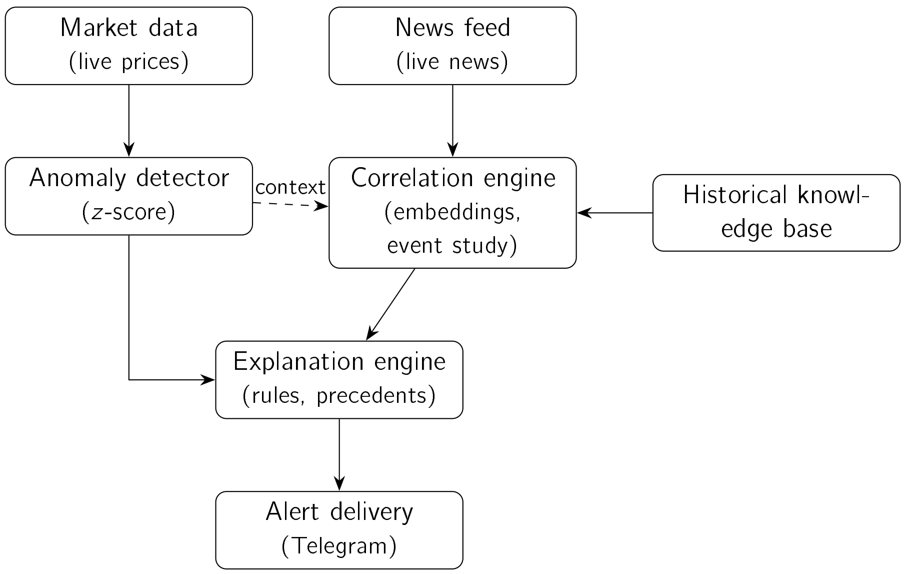

Figure 4.1: Architecture of Clarion. Solid arrows show the data flow; the dashed arrow shows the anomaly providing context to the correlation engine. The two triggers (market movement via the anomaly detector; news via the correlation engine) converge on the explanation engine before delivery.

interchangeable implementations — a dependency-free lexical baseline and the real Sentence-BERT model — so that one can be substituted for the other without changing anything else, which is what makes the embedding-model comparison in Chapter 5 a fair one.

4.3 Historical Knowledge Base and Correlation Engine

The historical layer is a knowledge base of past news, each entry storing its date, ticker, headline, sentence embedding and measured post-event impact. Each news item is aligned to the market by taking the first trading day on or after the news date and measuring impact only from that day's close onward; this is the no-lookahead rule made concrete, and it avoids crediting the system with a jump already embedded in the opening price. The intended primary source is the FNSPID corpus (Dong, Fan, and Peng 2024); because building the full multi-year corpus is a long one-off job, the evaluation in Chapter 5 uses an initial knowledge base built from real recent news through the identical build process, and the full build is identified as future work.

The correlation engine has two parts. Similarity is computed with cosine over the (normalised) embeddings, returning the nearest historical items to a query. Impact is computed as the cumulative return at each horizon (+1, +3, +5 days) measured from the event close, with horizons that fall outside the available series reported as not available rather than silently dropped. Given a new item, the engine embeds it, ranks the knowledge base by similarity, and returns the top precedents with their scores — the evidence later shown to the user.

4.4 Anomaly Trigger

The anomaly trigger applies Equation 3.1 directly. For the day under evaluation it computes the mean and standard deviation over the window of strictly preceding days, forms the z-score of the latest return, and returns the decision together with every quantity that produced it — the z-score, window, threshold, mean and standard deviation. Exposing these values is what allows the explanation engine to be faithful by construction, and the evaluation applies exactly this rule across entire price series so that the experiments use the same logic as production.

4.5 Explanation Engine

The explanation engine renders alerts directly from the objects produced upstream, which guarantees that the text never diverges from the computation. For the market trigger, it states the move, the z-score, the threshold and window, and the recent mean and standard deviation. For the news trigger, it lists the retrieved precedents (date, ticker, similarity score, measured impact and headline) and the average impact across them, and it always ends with an explicit disclaimer that the figures are observed past outcomes, not a prediction — keeping the system within its no-forecasting scope.

4.6 Alert Delivery

Alerts are delivered to the investor through the Telegram messaging platform, chosen because it offers a free, widely used bot interface. The two end-to-end flows — retrieve, detect, explain and send for the market trigger; embed, retrieve precedents, explain and send for the news trigger — each encapsulate a complete run; periodic scheduling and deployment are deliberately left outside the present scope. A real alert was successfully delivered during testing, illustrated in Chapter 5.

4.7 Chapter Conclusions

This chapter described CLARION at the level of its architecture and design: a two-trigger, two-layer pipeline whose components are separated by responsibility and connected by a common schema, with explainability built in by rendering every alert directly from the system's own computation. The next chapter evaluates the system's two core components and its end-to-end behaviour on real data.

Chapter 5

Case Studies

This chapter evaluates CLARION through three case studies on real data, following the methodology fixed in advance in Chapter 3. The first studies the anomaly trigger across US equities of very different volatility; the second studies the news–market correlation engine and its retrieval of analogous precedents; the third walks through an end-to-end alert and examines the faithfulness of the explanation. Following the principle that modest but honest results are preferable to inflated ones, every number reported here is produced by a versioned script with a fixed random seed, and the limitations of each study are stated explicitly.

5.1 Common Experimental Setup

Two real datasets are used. For the anomaly trigger, daily closing prices for the fifteen large-capitalisation tickers listed in the data card were retrieved from a free market-data service over a three-year window, yielding 752 daily log-returns per ticker. For the correlation engine, 3,692 real financial-news headlines for the same tickers were collected through a free company-news service — the most recent month available on the free tier — spanning five sectors (technology, banking, energy, health and consumer staples). News headlines and prices are normalised to a common schema so that the live and historical layers are directly comparable. This recent, single-month window is sufficient for the retrieval experiment but is the reason a multi-year corpus (FNSPID) is left as future work for a full impact analysis. The no-lookahead, reproducibility and honesty guarantees of Section 3.6 apply to every study below.

5.2 Case Study 1: Transparent Anomaly Detection on US Equities

The question is whether the rolling z-score (Chandola, Banerjee, and Kumar 2009) is a better universal detector of abrupt moves than a naïve fixed-percentage threshold. The proxy label for a "true" anomaly is a daily move in the upper percentile of *that ticker*'s own return distribution ($|r| \geq$ the 99th percentile); the baseline is a single global threshold ($|r| \geq 3\%$) applied to every ticker.

Firing-rate consistency (main argument). The strongest evidence is independent of any label. A good universal detector should fire at a comparable rate across instruments; a volatility-blind threshold cannot. Table 5.1 and Figure 5.1 show that the fixed threshold fires on roughly 1% of days for a calm stock (KO) but on about 35% of days for a volatile one (TSLA, NVDA), whereas the z-score fires at a near-constant rate ($\approx 2\%$) for all tickers.

The range of firing rates across tickers is twenty times tighter for the z-score, confirming that normalising by recent volatility makes detection comparable across the market.

Table 5.1: Firing rate across the fifteen tickers (three years of daily data).

Method	Min rate	Max rate	Range
z-score (window 20, ± 3)	0.015	0.032	**0.017**
Fixed threshold ($\lvert r \rvert \geq 3\%$)	0.011	0.354	**0.343**

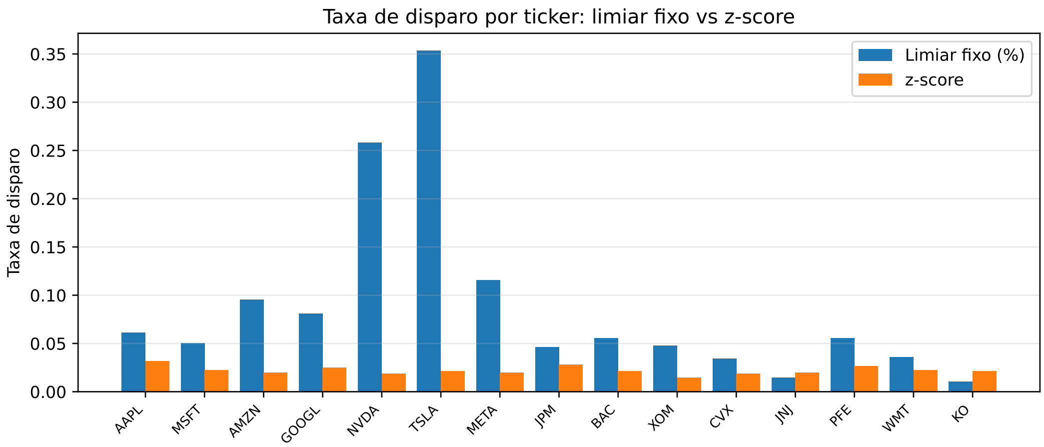

Figure 5.1: Firing rate per ticker. The fixed percentage threshold varies dramatically with each stock's volatility, while the z-score is stable across all tickers.

Agreement with the proxy label (supporting). Against the per-ticker extreme-move label, the z-score attains an F_1 of 0.524 (precision 0.388, recall 0.808) versus 0.216 for the fixed threshold (precision 0.121, recall 1.000): the fixed rule trivially catches every large raw move but at very low precision. A window ablation shows F_1 rising with the estimation window (0.385, 0.524 and 0.687 for 10, 20 and 60 days), indicating that a longer volatility estimate yields a steadier baseline. This label is itself volatility-relative, so some circularity is unavoidable; that is precisely why the firing-rate consistency above — which depends on no label — is treated as the primary evidence.

5.3 Case Study 2: News–Market Precedent Retrieval

The central claim of this work is that semantic retrieval finds genuinely analogous precedents. This is measured as precision@k with a sector proxy, in a *cross-ticker* setting: for each query the k most similar headlines *from other companies* are retrieved, and precision is the fraction sharing the query's sector. Excluding the query's own company prevents the trivial solution of matching on the company name and tests thematic analogy instead. To check that the result is not tied to one model, *two* Sentence-BERT models (Reimers and Gurevych 2019) — the lightweight default, MiniLM, and the larger MPNet — are compared against a lexical (hashing) baseline and two trivial baselines: random retrieval (the sector base rate) and recency. To guard against sampling artefacts, five hundred queries were sampled in each of five independent runs (seeds 42–46); Table 5.2 reports the mean and standard deviation.

Table 5.2: Cross-ticker precision@k by sector, mean $\pm$ std over five runs (3,692 real headlines, five sectors).

Method	**P@5**	**P@10**
SBERT — MiniLM (default)	0.549 ± 0.014	0.516 ± 0.013
SBERT — MPNet	$\mathbf{0.569 \pm 0.009}$	$\mathbf{0.538 \pm 0.009}$
Lexical baseline (hashing)	0.359 ± 0.010	0.327 ± 0.011
Recency	0.105 ± 0.013	0.081 ± 0.006
Random (base rate)	0.241 ± 0.004	0.241 ± 0.004

As Table 5.2 and Figure 5.2 show, the default MiniLM model reaches a precision@5 of 0.549 — about 2.3 times the random base rate (0.241) and well above the lexical baseline (0.359). The stronger MPNet model does slightly better (0.569), confirming that the advantage is a property of semantic embeddings in general and not of one particular model; the small standard deviations (around 0.01) confirm the gap is robust rather than a sampling artefact. Recency performs worst, showing that simply surfacing the latest news is no substitute for semantic matching. The lift over both the random and lexical baselines is the honest measure of the value added by the embeddings.

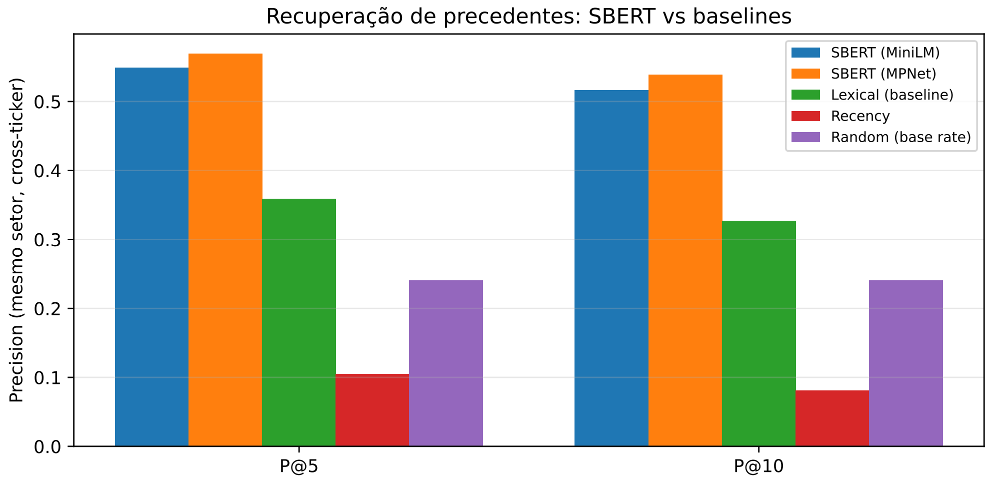

Figure 5.2: Cross-ticker precision@k by sector. SBERT clearly exceeds the lexical, recency and random baselines.

Impact measurement. For each retrieved precedent the engine reports the observed post-event return at +1, +3 and +5 trading days, measured from the event-day close in the style of an event study (Brown and Warner 1985). These figures are observed outcomes presented as evidence, never a prediction. On the worked knowledge base the measured impacts are coherent with reality — for example, a price-cut headline for one carmaker is followed by a sizable negative three-day return, and a cloud-growth earnings headline by a clear positive one — which provides face validity for the measurement; a rigorous multi-year impact study on the full FNSPID corpus is identified as future work.

5.4 Case Study 3: End-to-End Alert and Explanation Faithfulness

To illustrate the news trigger end to end, consider a new headline for NVIDIA, *"Nvidia raises guidance on surging demand for AI data-centre accelerators"*. The system embeds it, retrieves the most similar precedents from the knowledge base built from the real news corpus, and renders the alert below (top five precedents, one-day horizon):

```
News alert for NVDA
"Nvidia raises guidance on surging demand for AI data-centre
accelerators"
Potential impact (from 5 similar past events): average 1-day move:
−1.78%
Historical precedents:
- 2026-06-16 META (sim 0.64) +1d −5.44%: "Nvidia Latest Big
Borrower: AI Data Centers..."
- 2026-06-01 BAC (sim 0.63) +1d +1.88%: "Nvidia expands AI platform
with new CPU and PC..."
- 2026-06-18 AMZN (sim 0.61) +1d n/a: "Amazon hopes to challenge
Nvidia by selling AI chips"
- 2026-06-18 AMZN (sim 0.61) +1d n/a: "Amazon's New Weapon: Selling
Custom AI Chips..."
- 2026-06-18 AMZN (sim 0.60) +1d n/a: "Amazon explores selling AI
chips to challenge Nvidia"
Note: precedents are retrieved by semantic similarity; the impact is
the observed past outcome, not a price prediction.
```

Every retrieved precedent is thematically about NVIDIA and AI accelerators, even though the articles were filed under different company feeds (Meta, Bank of America, Amazon). This is direct evidence that retrieval is driven by *meaning* rather than the company name or exact keywords — the very behaviour the correlation engine is designed for. Where a precedent is too recent for its forward window to fall within the available prices, the impact is shown as not available; this is an honest artefact of the one-month free-tier window and further motivates the multi-year corpus. The alert ends with an explicit disclaimer, keeping the system within its no-forecasting scope.

Explanation faithfulness. Faithfulness is guaranteed by construction: the alert text is rendered directly from the same objects the system computes — the z-score, window and threshold for the anomaly path, and the exact retrieved precedents with their similarity scores and measured impacts for the news path — so the explanation cannot diverge from the underlying computation. This is checked programmatically by an automated test that asserts the rendered alert reproduces exactly each retrieved precedent's date, ticker and similarity score and introduces none that were not retrieved; it is the kind of transparent, design-time explainability advocated in the XAI literature (Barredo Arrieta et al. 2020). Usefulness to a retail investor is inherently subjective; a small rubric-based human assessment (clarity, completeness, actionability) is planned but has not yet been conducted, and is therefore reported as a limitation rather than a result.

5.5 Discussion and Threats to Validity

The three case studies support the core design: volatility-normalised detection is consistent across the market where a fixed threshold is not, semantic retrieval surfaces markedly more sector-analogous precedents than lexical or trivial baselines, and the end-to-end alert is faithful to its own computation. All results rest on real data and are reproducible from versioned scripts.

Several threats to validity are stated honestly. The sector label is an automatic *proxy* for relevance, not a human judgement; the news corpus is the recent window available from a free service rather than the multi-year FNSPID history, and headlines are short, which bounds the semantics that can be captured; the anomaly proxy label is volatility-relative, so the firing-rate argument (label-free) is given primacy; and, although the retrieval result is averaged over five runs, no human usefulness study has yet been conducted. Crucially, and by design, the system performs no price prediction: it does not forecast returns or evaluate trading profit (out of scope). Addressing these points — the full FNSPID corpus, a human relevance and usefulness study, and a multi-year impact analysis — defines the main avenues for further work.

5.6 Chapter Conclusions

Across three case studies on real data, CLARION's anomaly trigger fired far more consistently than a fixed threshold, its correlation engine retrieved sector-analogous precedents well above every baseline and independently of the embedding model, and its explanations were faithful to the underlying computation by construction. The next chapter draws these findings together against the research questions.

Chapter 6

Conclusions

This dissertation set out to design, implement and evaluate an explainable, reproducible financial-alert system for US retail investors, driven by two independent triggers and delivered through Telegram. This chapter summarises what was achieved against the research questions, revisits the contributions, and states the limitations and the directions they open.

6.1 Main Conclusions

The work is best summarised by returning to the three research questions of Chapter 1.

RQ1 — transparent detection of abrupt movements. The answer is affirmative. A rolling z-score detector, fully transparent by construction, detects abnormal returns and — crucially — does so *consistently across instruments of very different volatility*, where a fixed-percentage threshold cannot. As reported in Chapter 5, the firing rate varied by only 0.017 across the fifteen tickers for the z-score versus 0.343 for the fixed baseline, and the z-score reached an F_1 of 0.524 against an extreme-move proxy, more than double the baseline. The method is simple enough that every alert can be explained to the investor.

RQ2 — analogous precedents without lookahead. The answer is affirmative for retrieval. Semantic retrieval with Sentence-BERT recovered markedly more sector-analogous precedents than every trivial alternative — a cross-ticker precision@5 of 0.549 ± 0.014 (over five runs) against 0.359 for a lexical baseline, 0.241 for random and 0.105 for recency — and the end-to-end case study showed it surfacing genuinely topical precedents across different company feeds. Impact is measured strictly after the event, in the style of an event study, so no future information leaks into the evidence. The measurement machinery is in place and behaves coherently; a large-scale, multi-year validation of impact magnitudes remains for future work.

RQ3 — faithful and useful explanations. Faithfulness is achieved: because each alert is rendered directly from the same objects the system computes (the z-score and its parameters, or the retrieved precedents with their scores and measured impacts), the explanation cannot diverge from the underlying logic, and every alert closes with an explicit no-prediction disclaimer. Usefulness to a retail investor is plausible by design but has not yet been measured with a human study, and is therefore reported as an open point rather than a settled result.

6.2 Research Questions and Objectives Achieved

The general objective — to design, implement and evaluate an explainable, reproducible alerting system driven by two triggers — was met: CLARION is a working system whose two triggers were proven end to end and whose two core components were evaluated on real data. The supporting objectives were also met: a thin end-to-end slice was delivered first; each component was kept in its simplest defensible form; the evaluation followed a design fixed in advance; and the whole system is reproducible from versioned scripts. RQ1 and RQ2 are answered affirmatively on real data, while RQ3 is answered for faithfulness and left partly open for usefulness, pending a human study.

6.3 Contributions Revisited

As an Artificial Intelligence *Engineering* dissertation, the contribution is the integration, application and critical evaluation of existing components into a working, explainable and reproducible whole, rather than a new algorithm. Three concrete contributions stand. First, a documented and reproducible methodology for **news–market impact correlation** that combines sentence-embedding retrieval with event-study impact windows, with explicit choices of similarity metric and horizon and explicit avoidance of lookahead. Second, **CLARION**, an **XAI-first alerting system** whose entire reasoning chain — detection, evidence, sources and precedents — is exposed to a non-expert user and is faithful by construction. Third, a **critical, honest evaluation** of each core component against trivial and lexical baselines on real data, with results, ablations and limitations reported transparently and reproducibly from versioned scripts.

6.4 Limitations

Several limitations are acknowledged honestly. The evaluation used a recent, single-month news corpus from a free service rather than the multi-year FNSPID history, which bounds the impact analysis and means the retrieval figures, while real, are preliminary. The relevance proxy is sector agreement, an automatic stand-in for human judgement, and the anomaly proxy label is volatility-relative (hence the label-free firing-rate argument is given primacy). News is represented by short headlines only, which limits the semantics that can be captured. The retrieval result is averaged over five runs, but no human usefulness study has yet been conducted. Finally, and by design, the system performs no price prediction or trading, so profit-based questions are out of scope rather than unanswered.

6.5 Future Work

The limitations map directly onto future work. The most immediate step is to build the full multi-year knowledge base from FNSPID, for which the streaming pipeline is already in place, enabling a richer impact analysis with dispersion and direction-consistency statistics. A small human study, applying a clarity–completeness–actionability rubric to a set of alerts, would address the open question of usefulness (RQ3). The retrieval could be strengthened by embedding full article bodies rather than headlines and by comparing further embedding models, and the explanation could be optionally enriched with financial sentiment or feature attribution where it adds defensible value. Together these would turn a sound, honestly evaluated prototype into a more thoroughly validated system, without departing from its founding principle of defensible, transparent simplicity.

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Appendix A

Reproducibility

This appendix records the engineering detail needed to reproduce the system and its results, kept out of the main text so that the body reads as a dissertation rather than a software specification.

A.1 Environment

The system is implemented in Python 3.12, chosen for the maturity of its packages for the machine-learning stack. Dependencies are pinned and frozen in a lock file so the environment can be recreated across machines. The numerical and data handling rely on established open-source libraries; sentence embeddings use the Sentence-BERT framework on a CPU build of the underlying deep-learning library, installed from a dedicated index to avoid the much larger GPU download; figures are produced with a standard plotting library.

A.2 Repository Organisation

The code is organised as one package per component, mirroring the architecture of Figure 4.1: market data, news retrieval, anomaly detection, the correlation engine (similarity and event study), the historical knowledge base, the explanation engine, and alert delivery, with a small orchestration layer wiring them into the two end-to-end flows. Pure logic is separated from input/output, and heavy or networked dependencies are loaded lazily, so the numerical core is unit-tested without a network connection or the model stack. Secrets are read only from a local, ignored environment file and are never written to versioned files.

A.3 Tests and Reproducible Results

The system is covered by an automated test suite spanning the anomaly detector, the event study, similarity, the knowledge base, the news parser, the explanation engine and the evaluation modules, together with an offline end-to-end smoke test of the news trigger. Tests that require network access or heavy models are marked and excluded from the default run so that it is fast, deterministic and offline. All experimental results in Chapter 5 are regenerated by scripts with a fixed random seed, and the result figures are written directly into the thesis figure directory, closing the loop between code and document.

A.4 Data and Attribution

Large data and models are never versioned; they are recreated by the data scripts, and only small samples are committed. The FNSPID dataset (Dong, Fan, and Peng 2024) is used under its CC BY-SA 4.0 licence, with attribution as required; the live market-data and news services are used within their free tiers.